

CS & IT ENGINEERING



Digital Logic

Combinational Circuits

Lecture No. 04



By- Aditya sir

Recap of Previous Lecture



Topic

Intro to mux



Multiplexed

mux

Set 1



Topics to be Covered



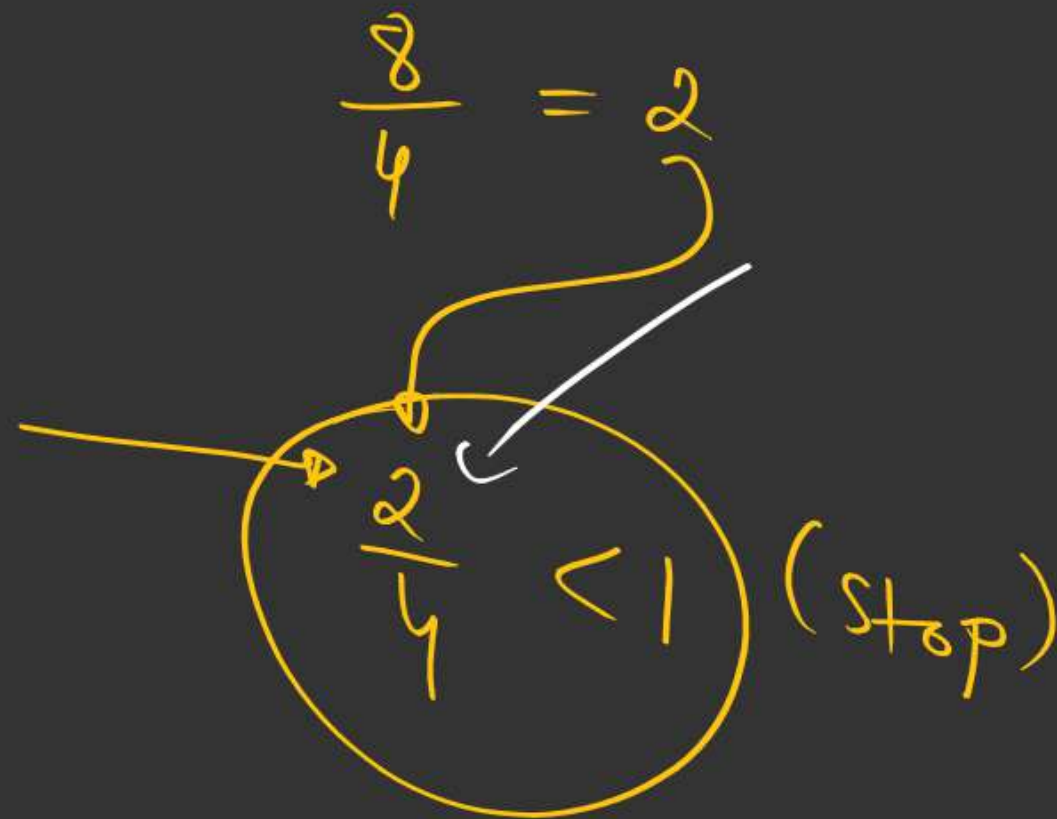
Topic



Diff types of Problems
on MUX.

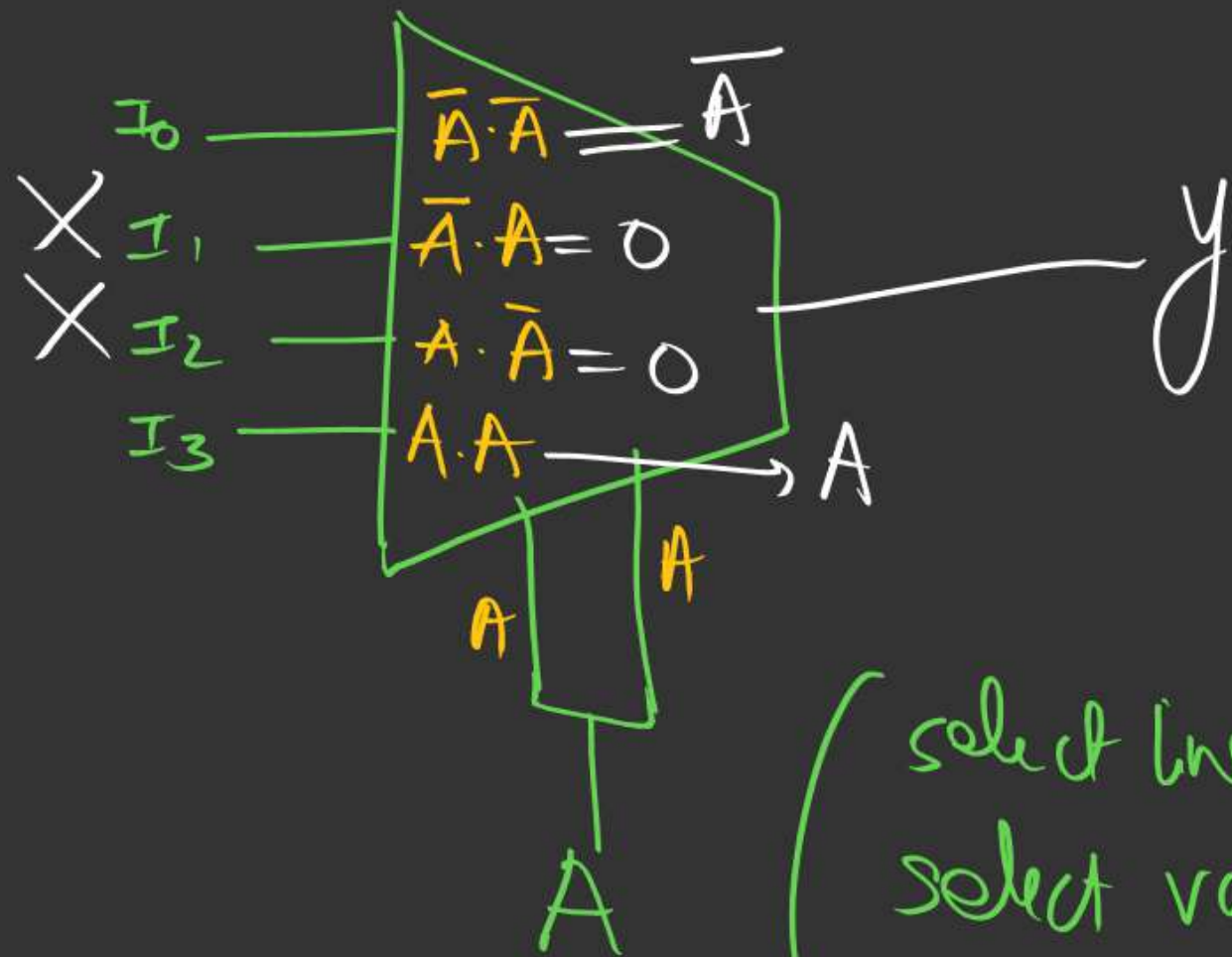
Imp

Create 8x1 using 4x1?

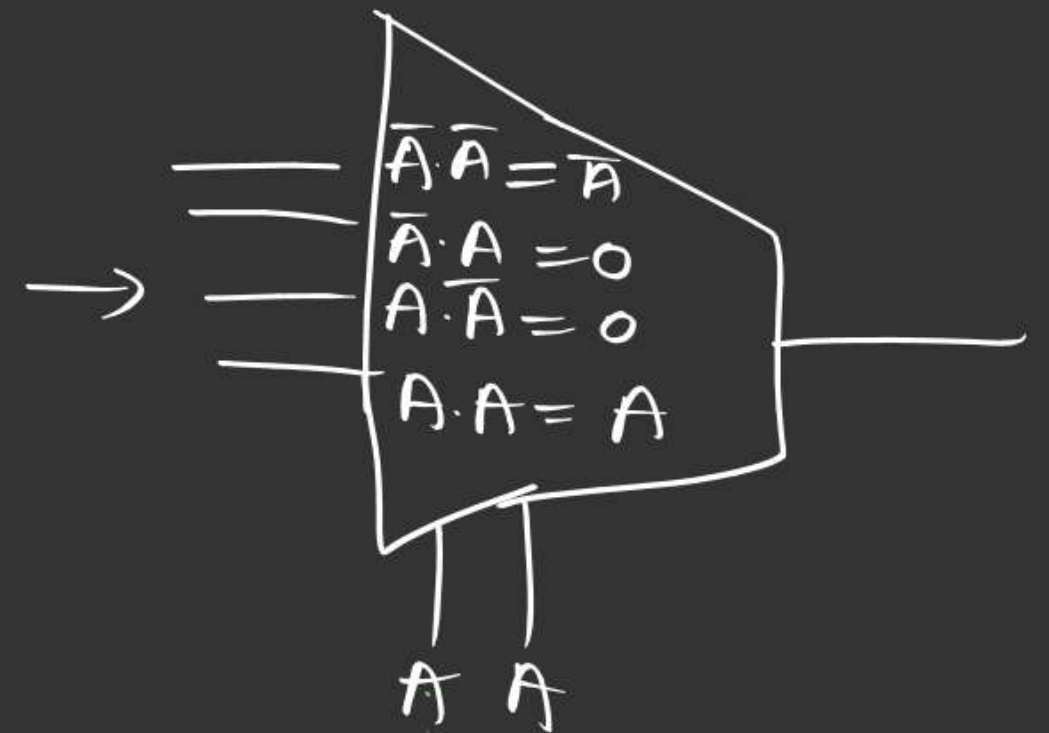


$$\text{Total} = 2 + 1 = \underline{\underline{3}} \checkmark$$

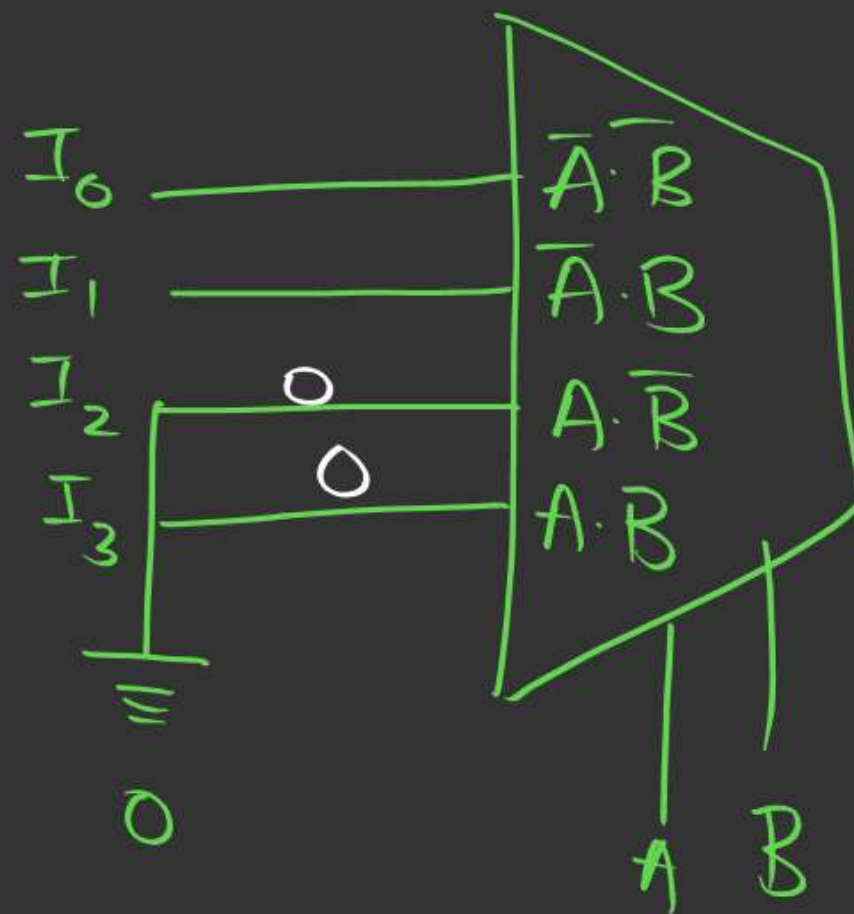
✓ Approach 1:-



(select lines = 2
select variable = 1)

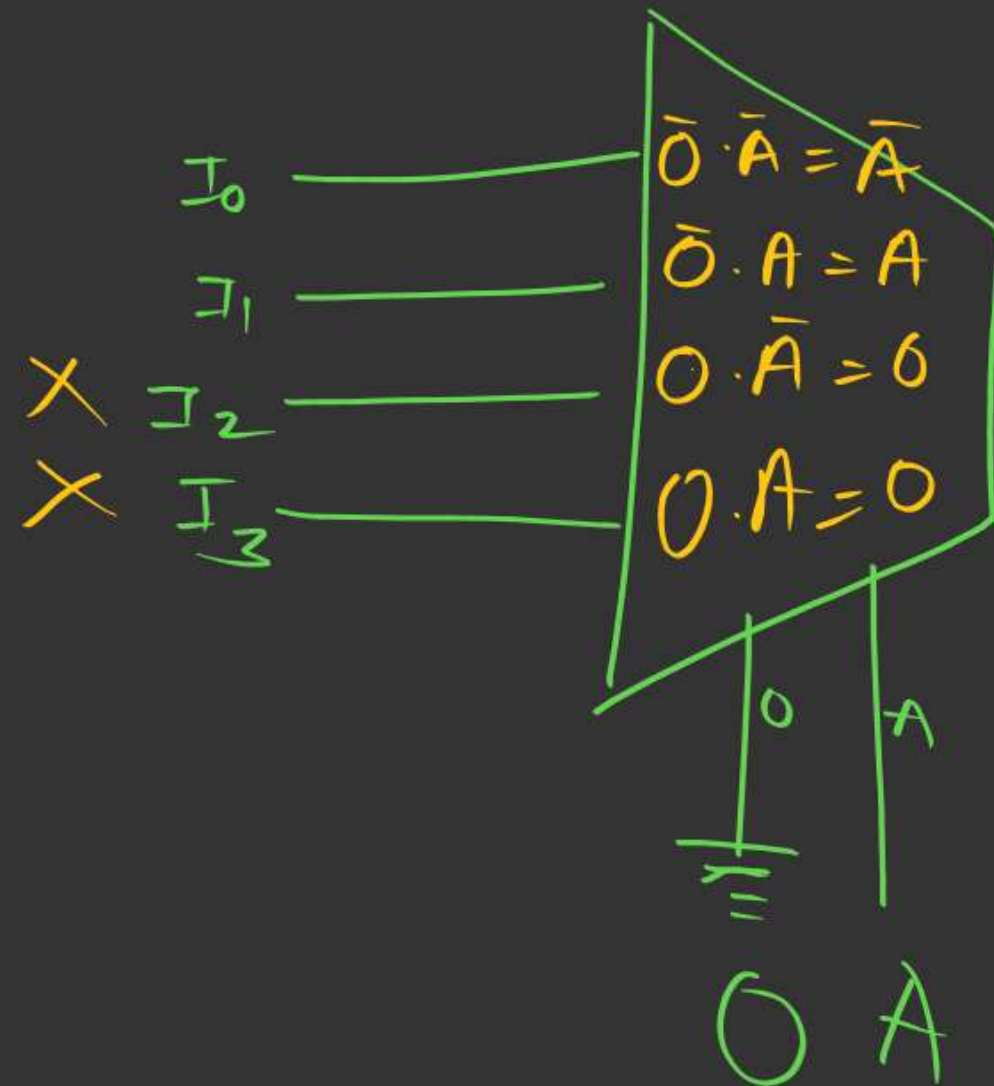


Appr 2 :-



(2 select variable)

Appr 3:-



(select variables = 1)

Design using Appx 3:

S_2, S_1, S_0

8x1
 $\downarrow$
3SL

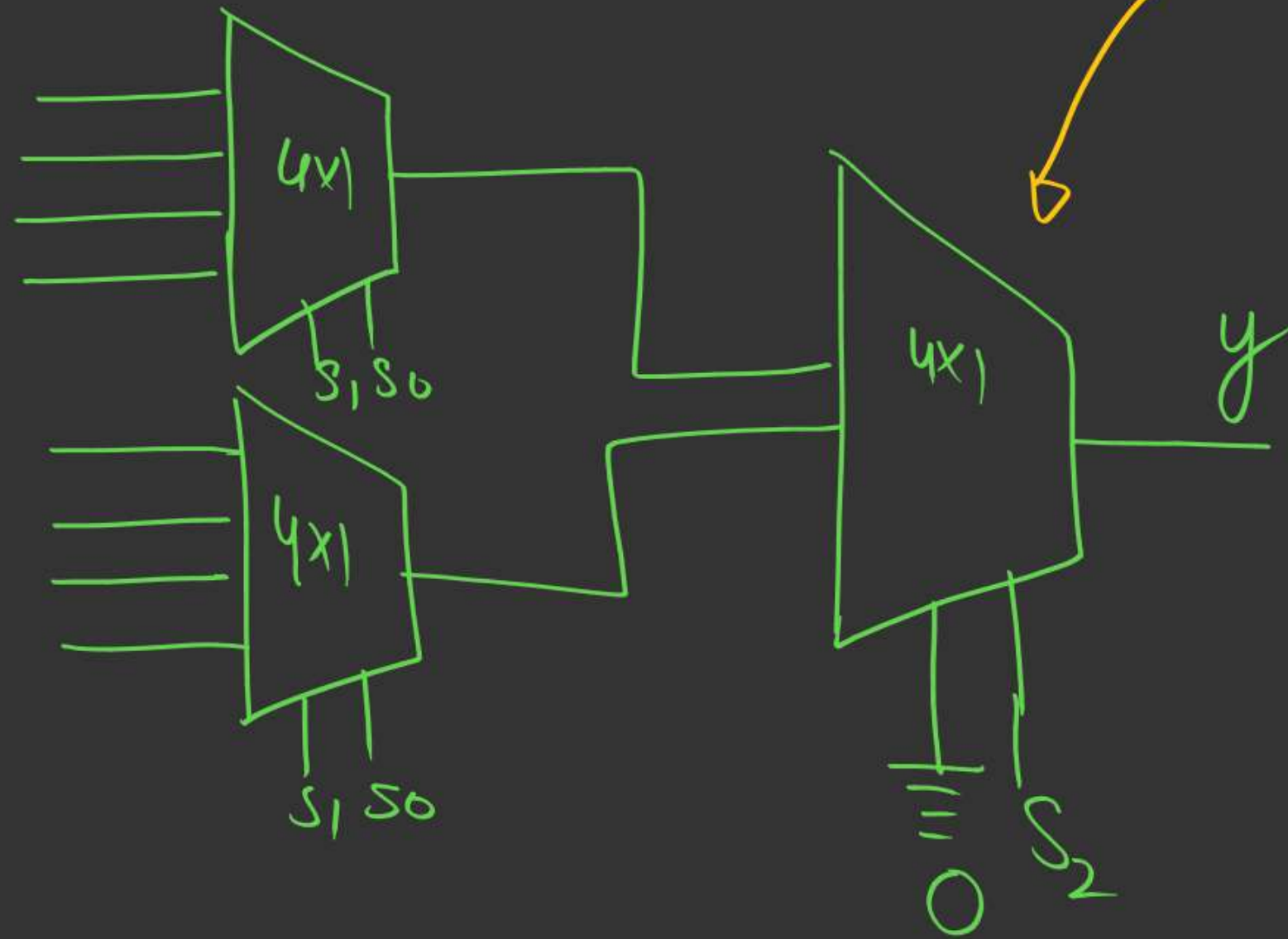
using 4x1
 $\downarrow$
2SL

$$\frac{8}{4} = 2$$

$$\frac{2}{4}$$

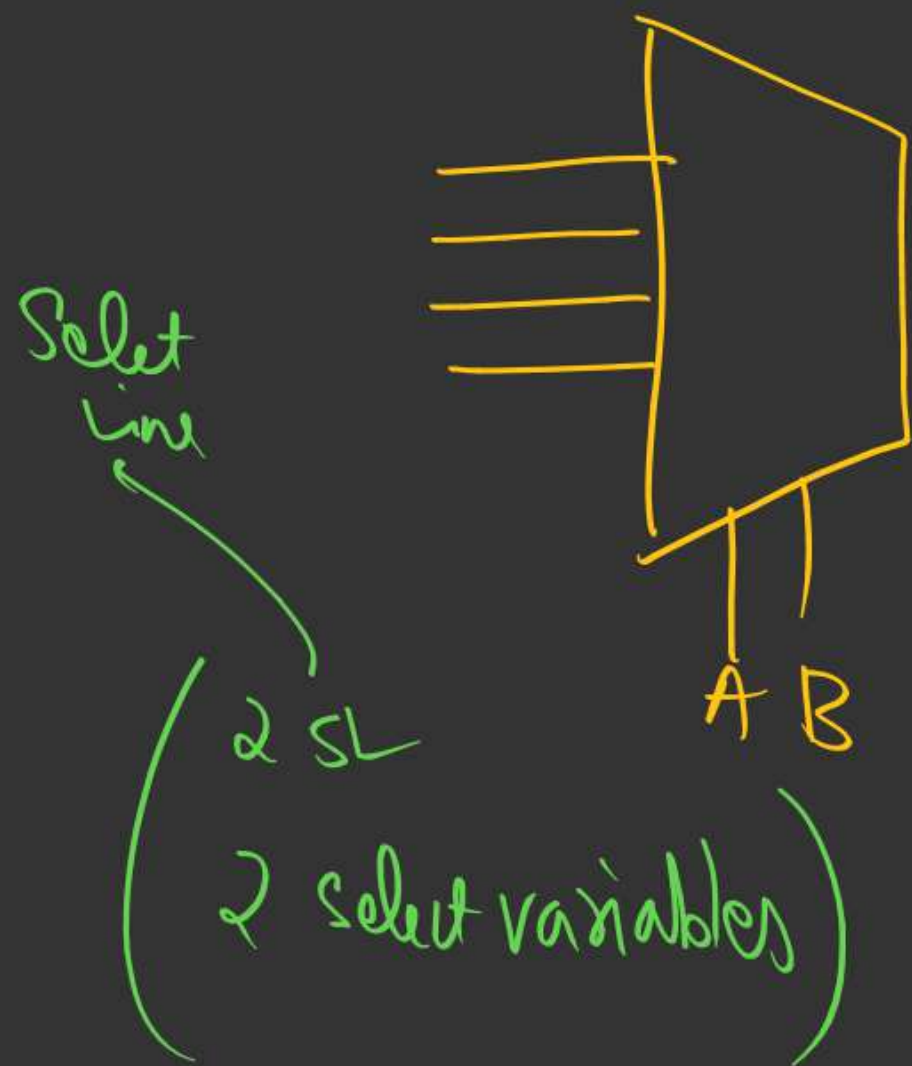
$$< 1 \Rightarrow 1$$

$$(2+1=3 \text{ Total})$$



3 (4x1) mux

select line vs select variable.



(2 SL
1 select variable)

How many 4×1
to design 32×1 ?

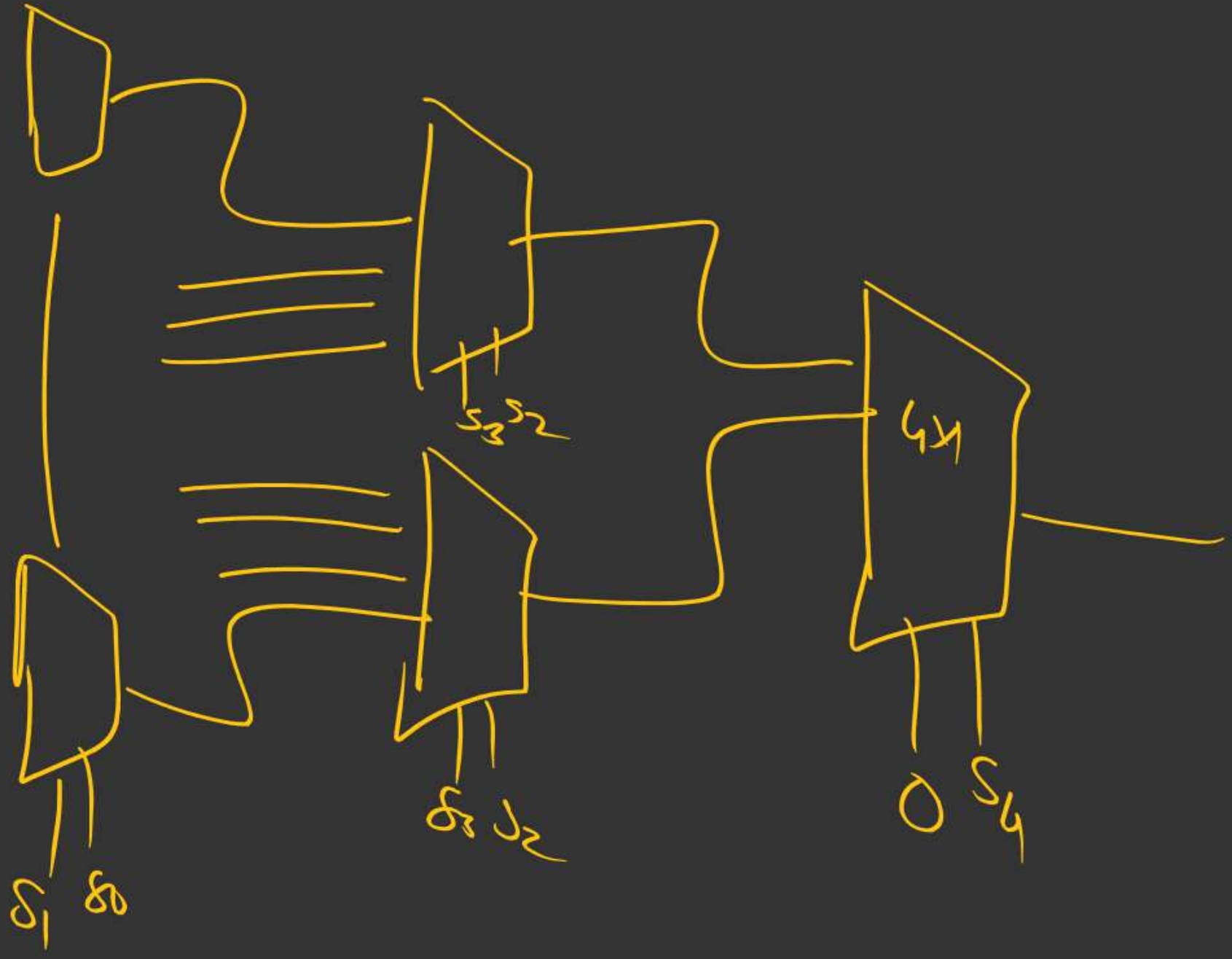
→ Design outline

$$\frac{32}{4} = 8$$

$$\frac{8}{4} = 2$$

$$\frac{2}{4} < 1$$

8



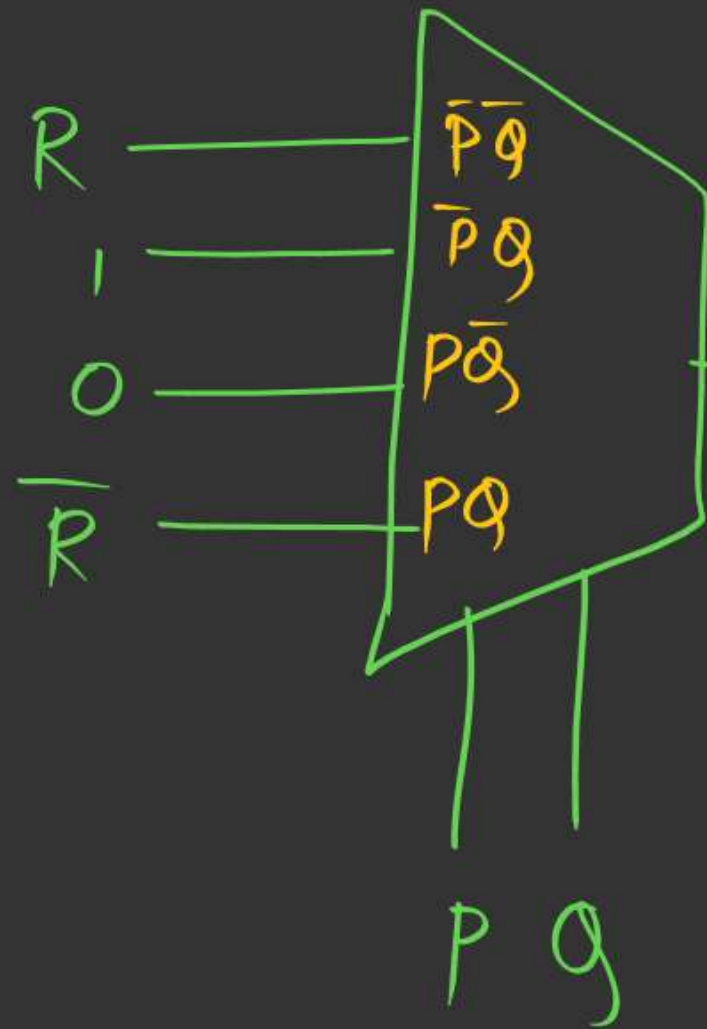
$8 + 2 + 1 = \textcircled{11}$ 4x1 mux

Set 2 Problems on MUX

↳ MUX as a universal logic (Gate)

Concept

eg:-



$$F(P, Q, R)$$

$$Y = \overline{P}\overline{Q}R + \overline{P}Q \cdot 1 + P\overline{Q} \cdot 0 + PQ\overline{R}$$
$$= [\overline{P}\overline{Q}R + \overline{P}Q + P\overline{Q}\overline{R}]$$

mux

→ NOT

→ OR

→ AND

→ NAND

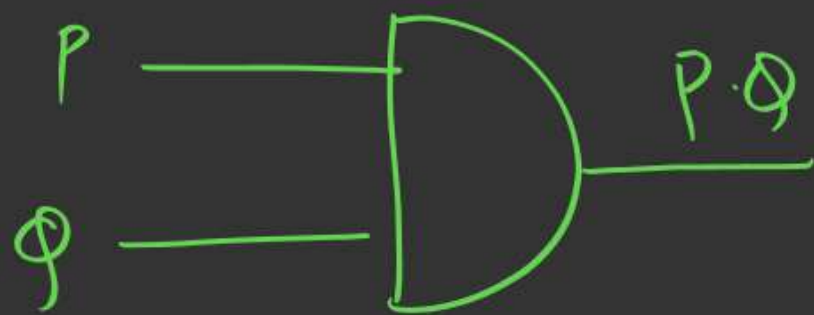
→ NOR

→ XOR

→ XNOR

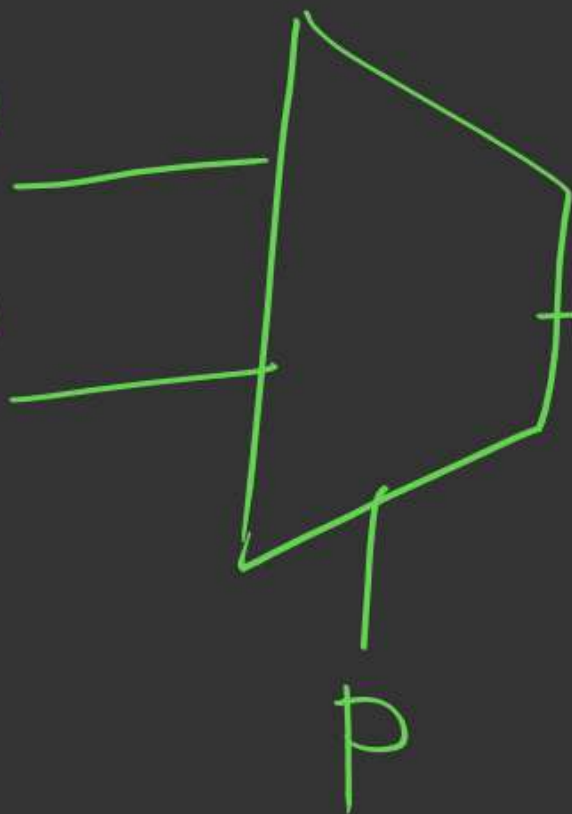
for proof: 2x1 mux

✓ 2) AND gate



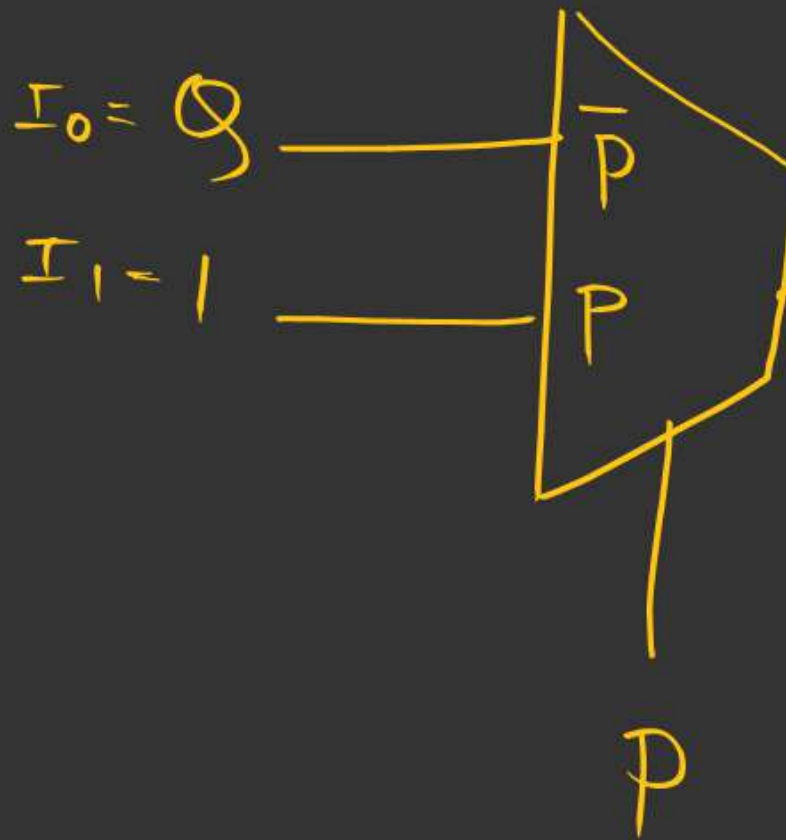
$$I_0 = 0$$

$$I_1 = Q$$



$$\begin{aligned} Y &= \bar{P} \cdot I_0 + P \cdot I_1 \\ &= \bar{P} \cdot 0 + P \cdot Q \\ &= \underline{\underline{P \cdot Q}} \end{aligned}$$

✓ 3 OR gate



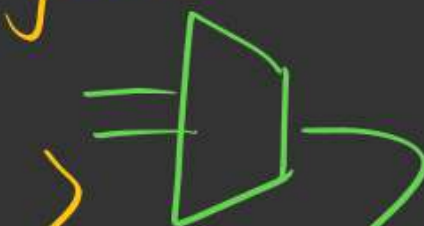
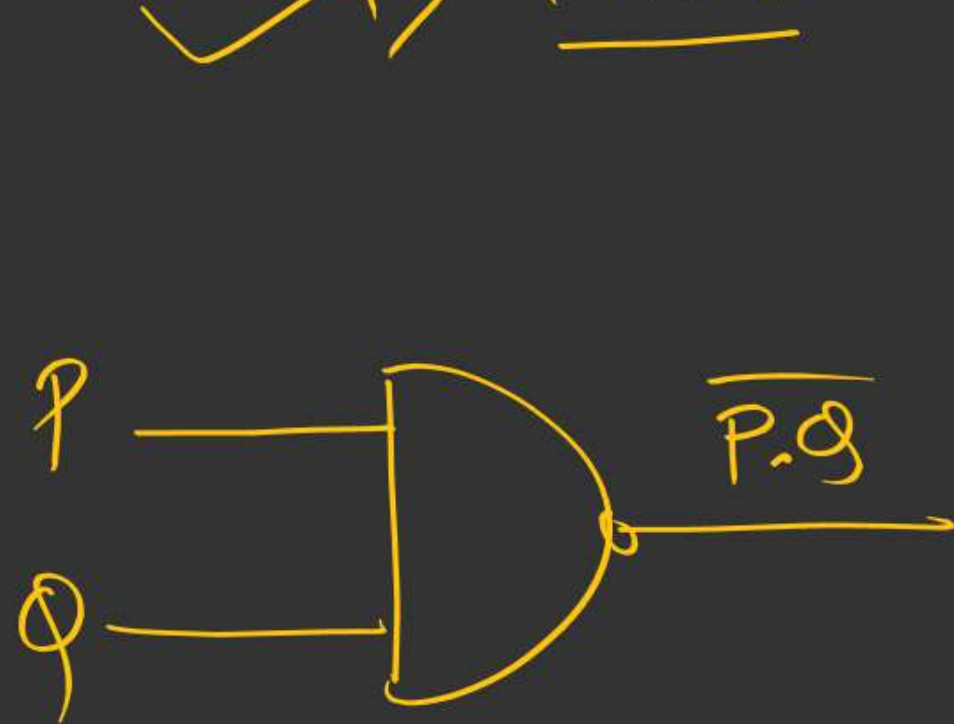
$$y = \bar{P} \cdot Q + P \cdot 1$$

$$= P + \bar{P} Q$$

$$= (P + \bar{P})(P + Q)$$

$$= \underline{\underline{P + Q}}$$

✓ NAND gate



$$I_0 = 1$$
$$I_1 = \overline{Q}$$



$$y = \overline{P} \cdot 1 + P \cdot \overline{Q}$$

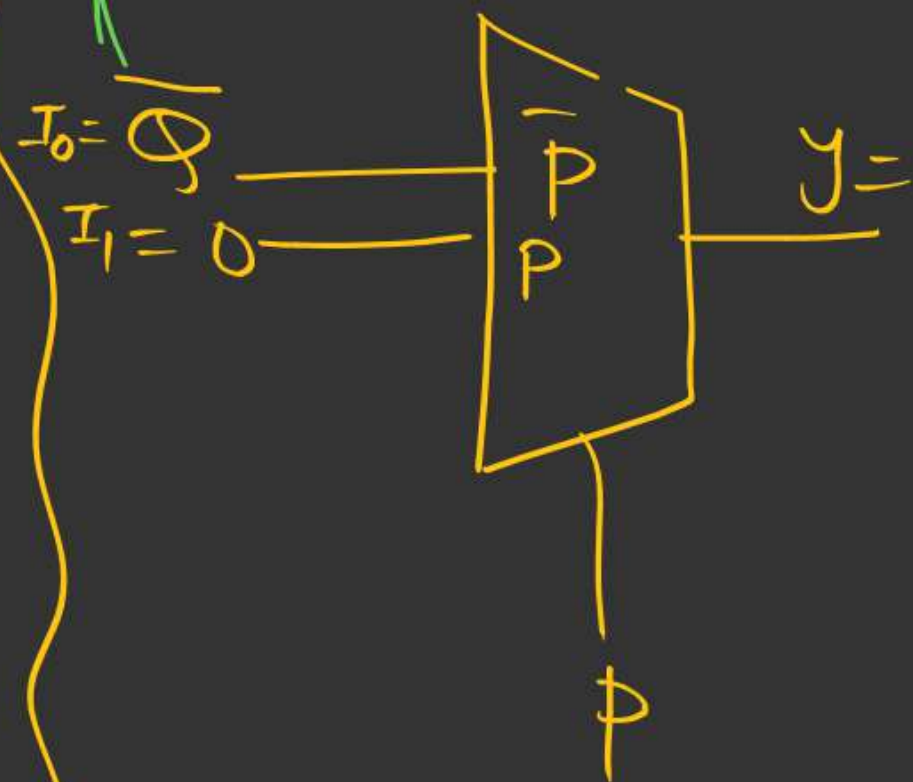
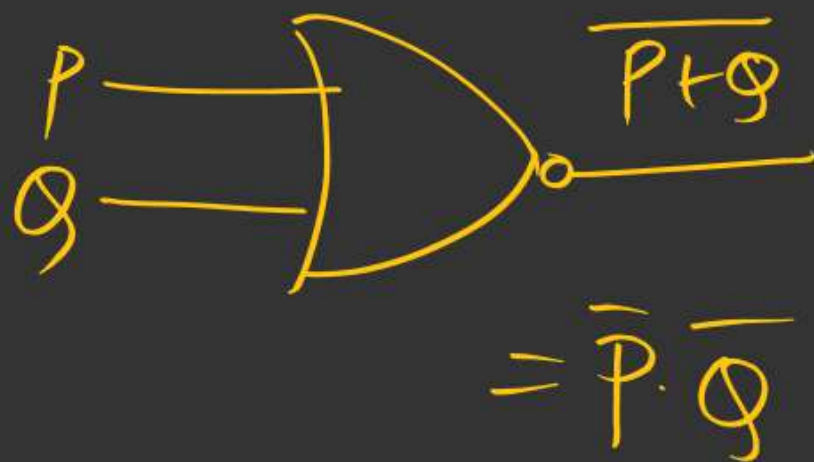
$$= \overline{P} + P \cdot \overline{Q}$$

$$= (\overline{P} + P)(\overline{P} + \overline{Q})$$

$$= \overline{P} + \overline{Q}$$

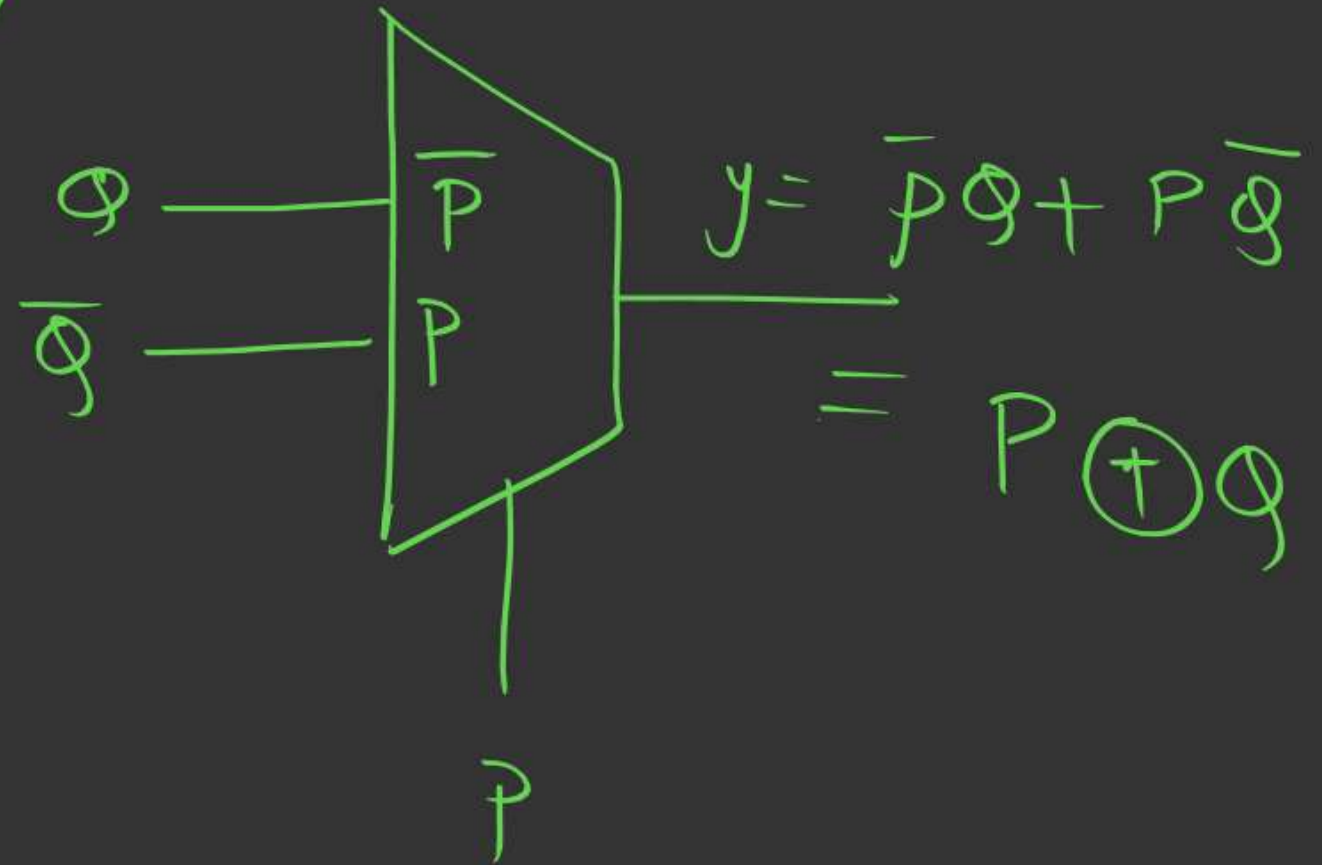
$$= \overline{P \cdot Q}$$

✓ NOR gate $\equiv \square$

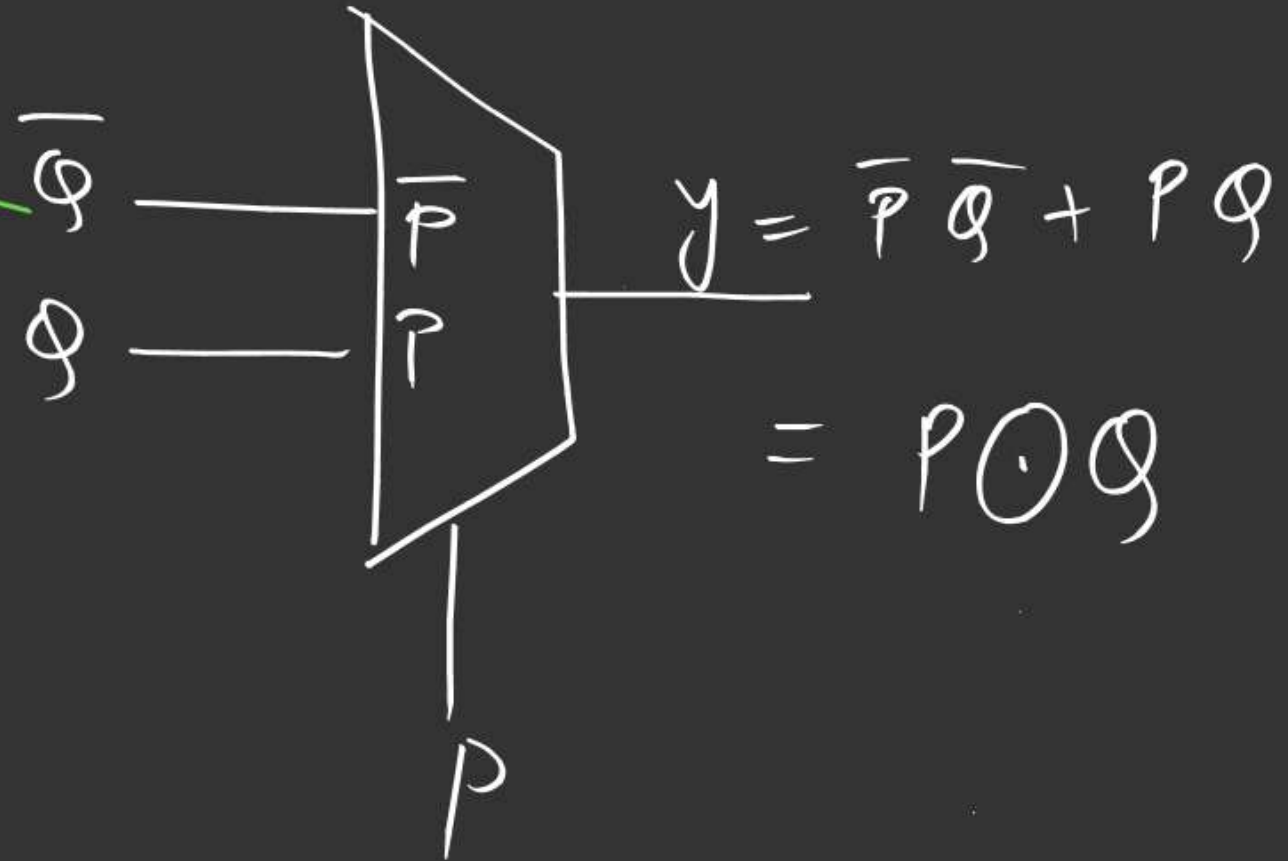
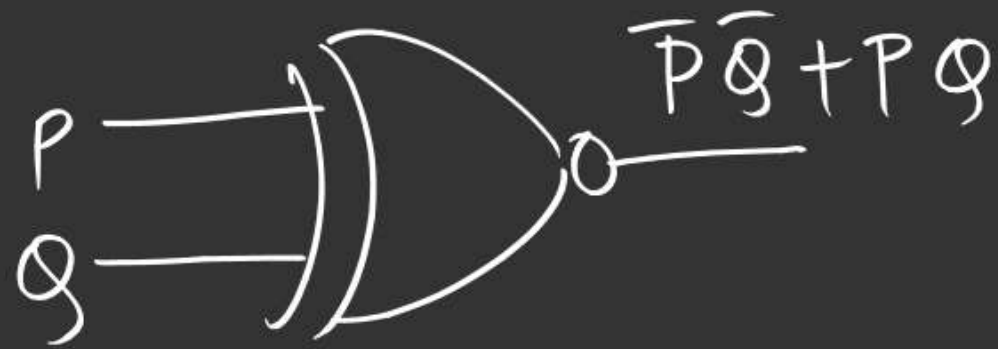


$$\begin{aligned} Y &= \bar{P} \cdot \bar{Q} + P \cdot 0 \\ &= \bar{P} \cdot \bar{Q} \\ &= \overline{P+Q} \end{aligned}$$

XOR gate



7 X-NOR gate



(Q) How many mux to design each type of gate.

2x1 mux

no
Compliment
/ps

- 1) NOT $\rightarrow$ 1
- 2) OR $\rightarrow$ 1
- 3) AND $\rightarrow$ 1
- 4) NAND $\rightarrow$ 2
- 5) NOR $\rightarrow$ 2
- 6) XOR $\rightarrow$ 2
- 7) XNOR $\rightarrow$ 2

Set 3 Questions on MUX



Minimization

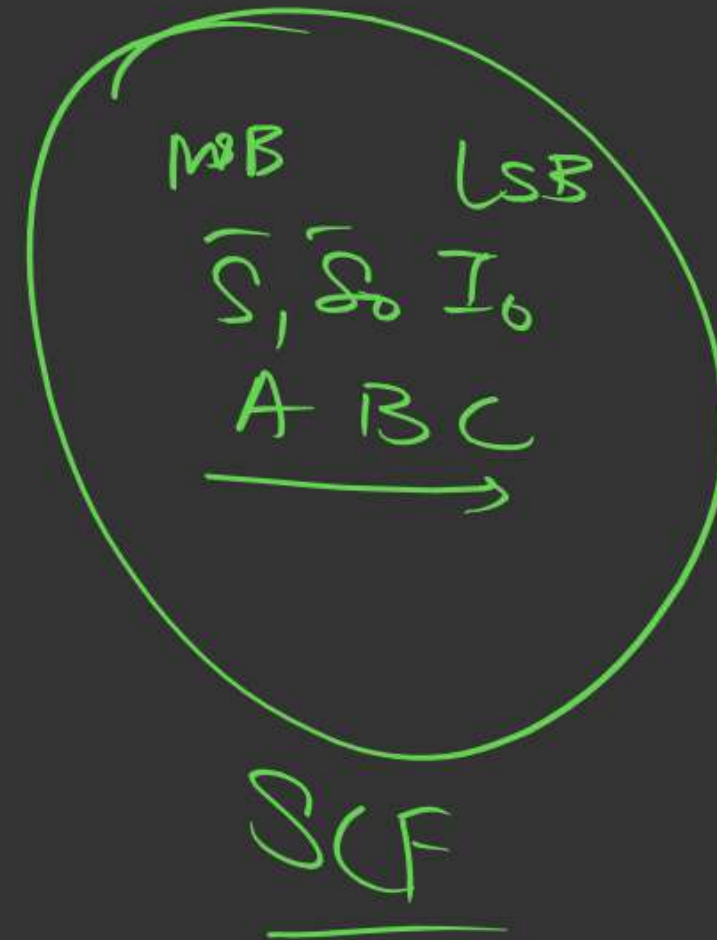
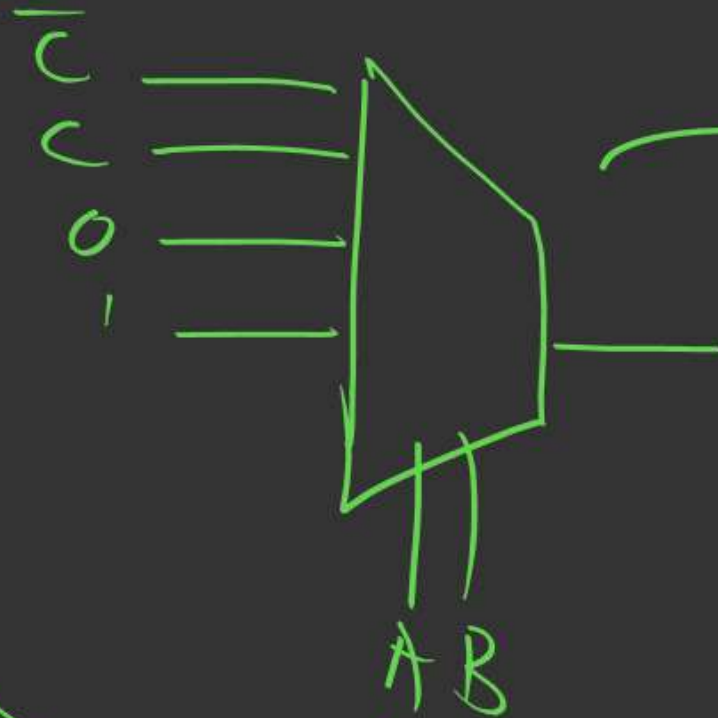
(MUX + K-map)

- Step 1: Find expression from MUX
- Step 2: Re-write expression in standard canonical SOP form.
- Step 3: minimize using K-map

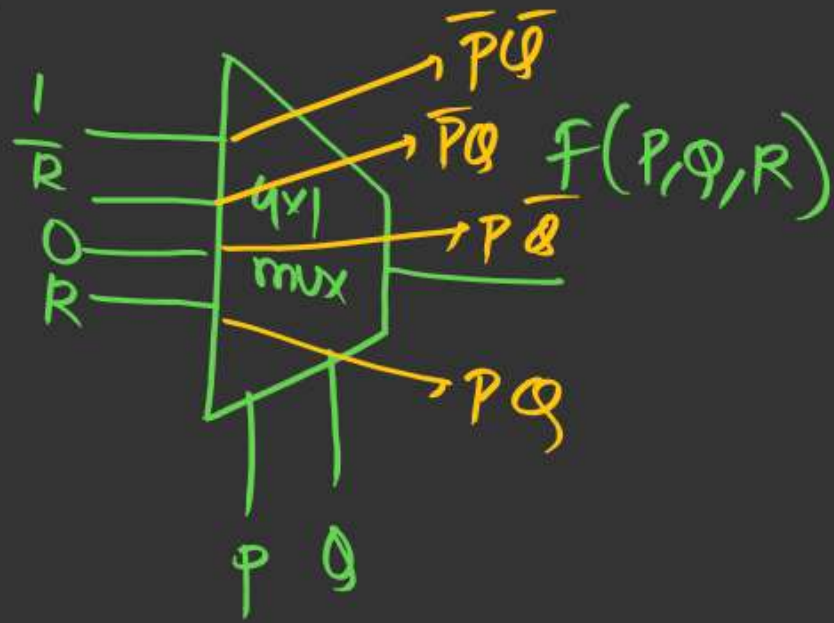
Ordering
Imp
for min-terms

$$\boxed{\begin{array}{l} \bar{S}_1, \bar{S}_0, I_0 = I_0 \bar{S}_0 \bar{S}_1 \\ 001 \neq 100 \end{array}} \leftarrow \text{SCF}$$

Imp: by default



eg1
Given find the minimized expression of $f(P, Q, R)$



Step 1: $f(P, Q, R) = \overline{P} \cdot \overline{Q} \cdot 1 + \overline{P} \cdot Q \cdot \overline{R} + P \cdot \overline{Q} \cdot 0 + P \cdot Q \cdot R$
 $= \overline{P} \overline{Q} + \overline{P} Q \overline{R} + P Q R$

Step 2:- SGF (SOP)

$$\rightarrow F(P, Q, R) = \bar{P}\bar{Q} + \bar{P}Q\bar{R} + PQR$$

$$= \bar{P}\bar{Q} \cdot (R + \bar{R}) + \bar{P}Q\bar{R} + PQR$$

$$= \bar{P}\bar{Q}R + \bar{P}\bar{Q}\bar{R} + \bar{P}Q\bar{R} + PQR$$

Step 3:-

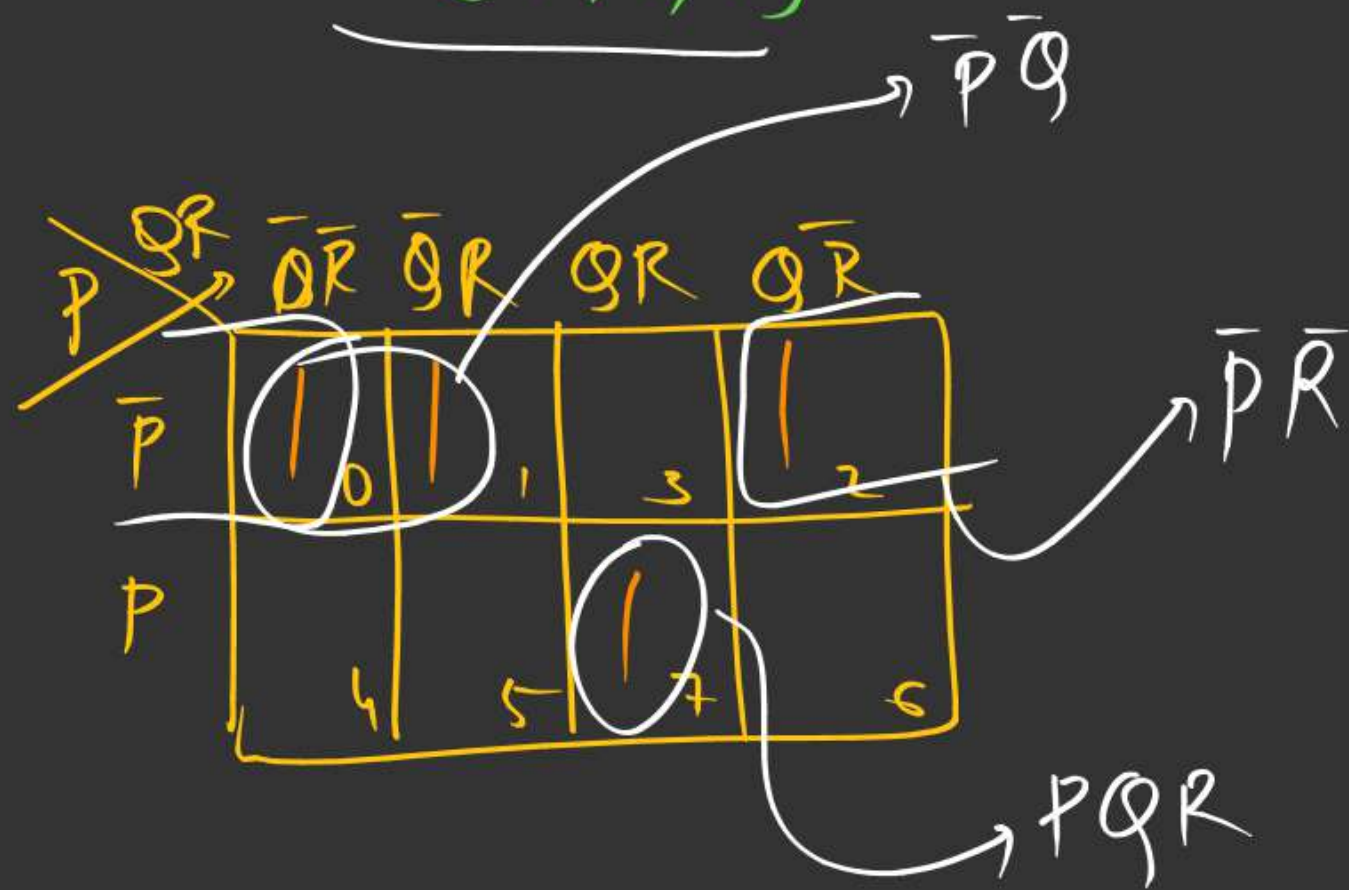
$$\bar{P}\bar{Q}R + \bar{P}\bar{Q}\bar{R} + \bar{P}QR + PQR$$

001, 000, 010, 111

↓ ↓ ↓ ↓
1 0 2 7

$\bar{P} \rightarrow 0$
 $P \rightarrow 1$

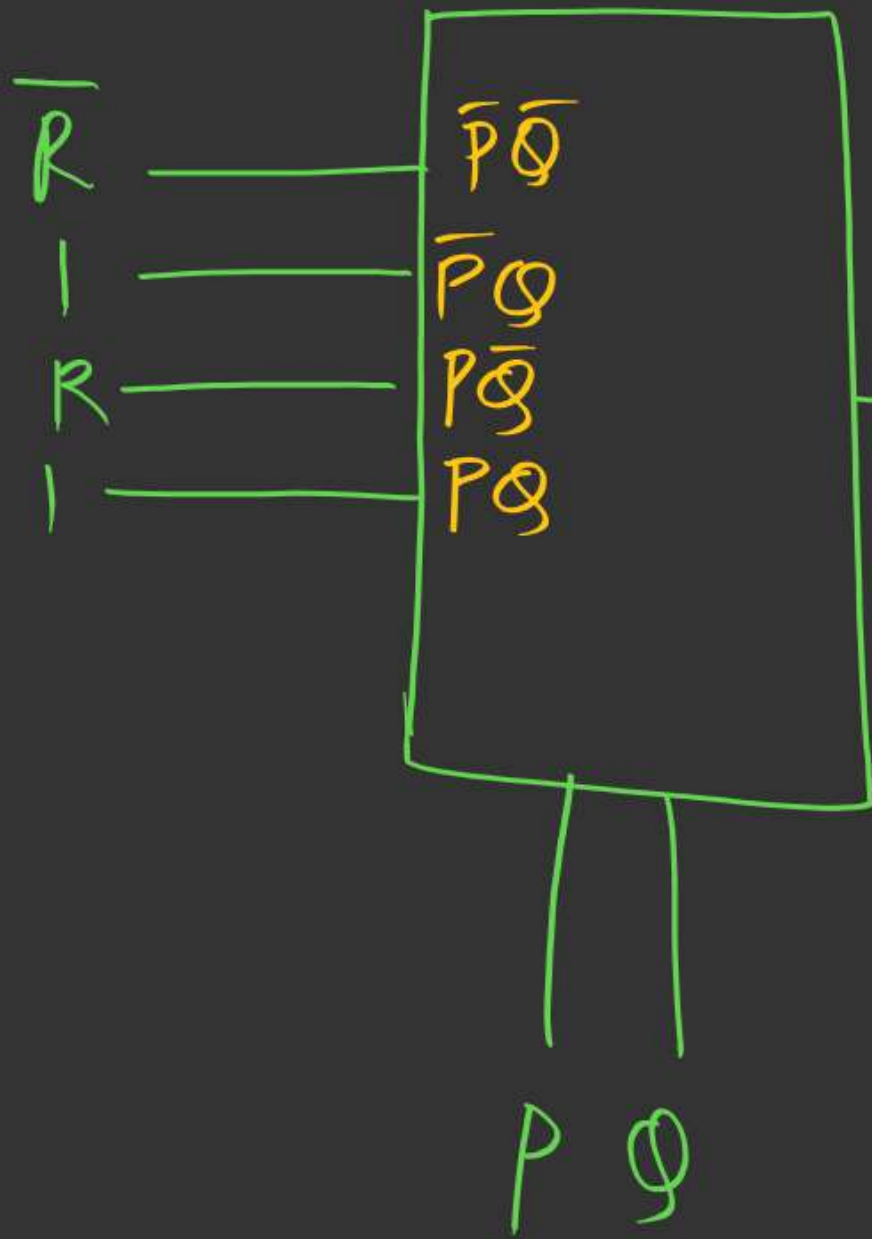
min terms $\Rightarrow F(P, Q, R) = \Sigma(0, 1, 2, 7)$



$$F(P, Q, R)$$

$$= [\bar{P}\bar{Q} + \bar{P}\bar{R} + PQR]$$

(Q) $F(P, Q, R) = ?$



$F(P, Q, R) = ?$

Step 1: $F(P, Q, R) = \overline{P}\overline{Q}\overline{R} + \overline{P}Q\cdot 1 + 1\overline{Q}\cdot R + P\overline{Q}\cdot 1$
 $= \overline{P}\overline{Q}\overline{R} + \overline{P}Q + P\overline{Q}R + P\overline{Q}$

Step 2:-

$$f(P, Q, R) = \bar{P}\bar{Q}\bar{R} + \bar{P}Q\bar{R} + P\bar{Q}\bar{R} + PQR$$

$$= \bar{P}\bar{Q}\bar{R} + \bar{P}Q + P\bar{Q}\bar{R} + PQR$$

$$= \bar{P}\bar{Q}\bar{R} + \bar{P}Q(R + \bar{R}) + P\bar{Q}\bar{R} + PQR$$

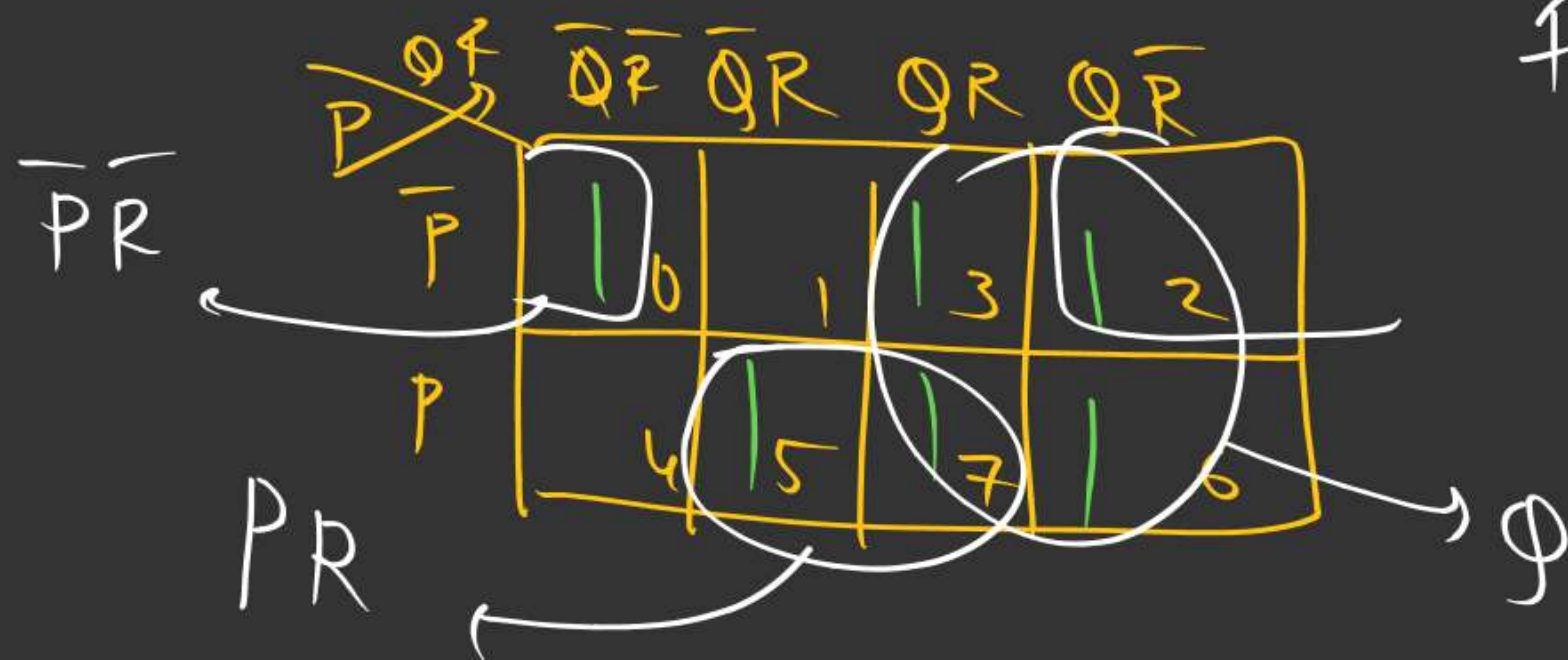
$$= \bar{P}\bar{Q}\bar{R} + \bar{P}QR + \bar{P}Q\bar{R} + P\bar{Q}\bar{R} + PQR + PQR$$

Step 3:-

$$= \bar{P}\bar{Q}\bar{R} + \bar{P}QR + \bar{P}Q\bar{R} + P\bar{Q}R + PQR + PQ\bar{R}$$

000	011	010	101	111	110
↓	↓	↓	↓	↓	↓
0	3	2	5	7	6

$$f(P, Q, R) = \sum (0, 2, 3, 5, 6, 7)$$



$$f(P, Q, R) = [\bar{P}\bar{R} + PR + Q]$$

$$= (P \odot R) + Q$$



2 mins Summary



set 2 ✓

set 3 → 50%



THANK - YOU

